

# FMCG Data Consolidation & Analytics Platform

End-to-End Data Engineering Pipeline using Medallion Architecture

Celebal Technologies — Summer Internship Project 2026

*Platform: Databricks (Lakehouse) | Language: PySpark | Storage: Delta Lake*

# 1. Introduction

A large FMCG retail company has recently acquired a smaller competitor. Both organizations operate with independent data systems, schemas, and storage formats. This has resulted in fragmented data across systems, making it difficult to derive unified insights. The absence of standardized data structures and centralized reporting has led to inefficiencies in decision-making and operational delays.

## 2. Business Problem

Post-acquisition, the organization is facing several data-related challenges:

- Data silos across multiple systems
- Inconsistent product, customer, and sales data
- Lack of unified reporting and analytics
- Delayed business decision-making

These challenges highlight the need for a scalable and efficient data platform that can integrate and standardize data across the enterprise.

## 3. Objective

The primary objective of this project is to design and implement an end-to-end data engineering pipeline that:

- Consolidates data from multiple heterogeneous sources
- Cleans, transforms, and standardizes datasets
- Supports both real-time and historical data analytics
- Enables business stakeholders with actionable insights through dashboards

## 4. Solution Overview (Lakehouse Architecture)

### 4.1 Data Ingestion (Bronze Layer)

Data is ingested from multiple source systems such as legacy databases, CSV files, and APIs. The raw data is stored in Amazon S3 (Data Lake) and processed using Databricks with PySpark. This layer preserves the original data for traceability and auditing purposes.

### 4.2 Data Transformation (Silver Layer)

In this layer, raw data is cleaned and standardized:

- Handling missing and null values
- Standardizing schemas across systems
- Removing duplicates and inconsistencies

Example: Merging customer IDs from both companies and normalizing product categories.

### 4.3 Business Logic (Gold Layer)

The transformed data is used to create business-ready datasets: fact and dimension tables are created, and KPIs and aggregations are applied.

Example Metrics: Total sales by region, Customer lifetime value, Product performance.

## 4.4 Data Modeling

A star schema is implemented:

- Fact Table: Sales transactions
- Dimension Tables: Customers, Products, Stores, Time

## 4.5 Incremental Processing

The pipeline supports historical data loads and incremental (daily/real-time) updates, ensuring efficient processing and scalability.

## 4.6 Orchestration

Workflow scheduling is implemented to automate the pipeline: data ingestion, transformation, and loading.

## 4.7 Visualization & BI

The final data is connected to BI tools such as Power BI and Databricks SQL dashboards, enabling stakeholders to track KPIs, monitor sales trends, and make data-driven decisions.

## 5. Technology Stack

| Component     | Technology                                    |
|---------------|-----------------------------------------------|
| Platform      | Databricks (Lakehouse Platform)               |
| Processing    | Apache Spark (PySpark)                        |
| Query Layer   | SQL                                           |
| Storage       | Amazon S3 (Data Lake) / Delta Lake            |
| Architecture  | Medallion Architecture (Bronze, Silver, Gold) |
| Visualization | Power BI / Databricks SQL Dashboards          |

## 6. Stakeholders

| Stakeholder       | Role                                 |
|-------------------|--------------------------------------|
| Business Analysts | Insights and reporting               |
| Data Engineers    | Pipeline development and maintenance |
| Management        | Strategic decision-making            |
| Sales & Marketing | Customer insights                    |

## 7. Pipeline Execution Results

### 7.1 Bronze Layer

| Source    | System             | Records |
|-----------|--------------------|---------|
| Company A | Superstore ERP     | 9,994   |
| Company B | Legacy FMCG System | 15      |
| Total     |                    | 10,009  |

### 7.2 Silver Layer

| Operation             | Result     |
|-----------------------|------------|
| Null values handled   | 301 values |
| Duplicates removed    | 8 rows     |
| Schema standardized   | Yes        |
| Final unified records | 10,001     |

### 7.3 Gold Layer (Star Schema)

| Table                       | Type      | Rows   |
|-----------------------------|-----------|--------|
| fact_sales                  | Fact      | 10,001 |
| dim_customer                | Dimension | 804    |
| dim_product                 | Dimension | 1,862  |
| dim_region                  | Dimension | 593    |
| dim_time                    | Dimension | 1,252  |
| kpi_sales_by_region         | KPI       | 8      |
| kpi_product_performance     | KPI       | 1,862  |
| kpi_customer_lifetime_value | KPI       | 804    |
| kpi_company_comparison      | KPI       | 2      |

## 8. Key Business Insights

### 8.1 Sales by Region

| Region  | Total Sales | Profit      | Orders |
|---------|-------------|-------------|--------|
| WEST    | Rs 7,13,369 | Rs 1,07,289 | 3,202  |
| EAST    | Rs 6,71,319 | Rs 91,434   | 2,845  |
| CENTRAL | Rs 4,97,800 | Rs 40,150   | 2,323  |
| SOUTH   | Rs 3,88,269 | Rs 46,450   | 1,616  |

### 8.2 Top Products

| Product            | Sales     | Profit    |
|--------------------|-----------|-----------|
| Canon imageCLASS   | Rs 61,599 | Rs 25,199 |
| Fellowes PB500     | Rs 27,453 | Rs 7,753  |
| Cisco TelePresence | Rs 22,638 | -Rs 1,811 |

### 8.3 Company Performance

| Company   | Total Sales  | Profit      | Orders |
|-----------|--------------|-------------|--------|
| Company A | Rs 22,70,758 | Rs 2,85,324 | 9,986  |
| Company B | Rs 39,400    | Rs 5,867    | 15     |

## 9. Business Impact

The implementation of this platform delivers significant value:

- Unified view of enterprise data
- Faster reporting and analytics
- Improved decision-making capabilities
- Scalable architecture for future growth
- Reduced data inconsistencies

*This project ensures a robust and scalable data foundation that supports business intelligence and long-term organizational growth.*

## 10. Conclusion

This project successfully demonstrates an end-to-end data engineering pipeline built on Databricks using Medallion Architecture. By consolidating heterogeneous FMCG data sources into a governed, business-ready Gold layer with a star schema, the platform enables faster, more reliable analytics and lays a scalable foundation for future enterprise data growth.